Language Models as Reads over Atom Spaces: A Unified Measure-Theoretic Formulation of Context, Retrieval, and Parameters

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Abstract

We reformulate language model next-token prediction as an integral of a query-conditioned probability measure against a value map over arbitrary atom spaces $(\Omega, \mathcal{F}, v)$, generalizing standard attention beyond token context windows. We demonstrate that in-context attention, retrieval-augmented generation (RAG / kNN-LM), and Mixture-of-Experts parameter routing are exact realizations of a single read operator $R(x; \Omega)$ applied to distinct atom spaces. This formulation decouples general linguistic reasoning from mutable factual storage, enabling post-hoc knowledge insertion without retraining. We establish the mathematical conditions governing reliable editability via key-encoder Lipschitz continuity and retrieval margins. Crucially, we prove the Quadrature Truncation Bound Theorem, showing that sublinear top-k Approximate Nearest Neighbor retrieval converges exponentially to the exact continuous integral under calibrated operational temperatures ($\tau = 0.05$, error $\approx 10^{-5}$). Extending reference measures to signed measures via the Hahn–Jordan decomposition, we prove that negative "anti-atoms" produce destructive interference, achieving exact uniform-entropy machine unlearning ($1/|\mathcal{V}| = 2.04\%$). Empirically, on a 200-fact technical benchmark, our architecture achieves 100% efficacy, generality, and specificity across density regimes and lifelong cumulative scaling up to 200 atoms. A learned convex router arbitrates across context, corpus, and parameter spaces with $> 92\%$ purity, where ablations confirm mutual necessity. Finally, integrating this operator into generative GPT-2 (124M) eliminates hallucinations, boosting target factual probabilities by up to 1100× and flipping top predictions to the correct target in 100% of cases.

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1. Introduction

Modern autoregressive language models conflate two fundamentally distinct computational roles into dense parameter matrices: general linguistic syntax and factual world knowledge [1, 2]. While gradient descent is well-suited for acquiring syntactic abstractions, semantic parsing, and structural reasoning, compiling static factual assertions directly into dense weights leads to severe practical bottlenecks. Factual knowledge is dynamic, highly localized, and subject to continuous obsolescence and legal erasure requirements (e.g., the GDPR "Right to be Forgotten"). When knowledge is frozen within billions of dense parameters, modifying or updating a single fact necessitates expensive retraining or fragile post-hoc weight editing [3, 4].

To circumvent the rigidity of dense parameter compilation, the community has developed disparate engineering solutions:

- *In-context reasoning* stores temporary information in the key-value cache of the sequence window [1].
- *Retrieval-augmented generation* (RAG) and nearest-neighbor language models (kNN-LM) query external vector databases of textual records [5, 6, 7].
- *Modular computation* mechanisms, such as Mixture-of-Experts (MoE) and parameter-efficient adapters (LoRA), route queries across discrete banks of specialized sub-networks [8, 9].

Although these techniques share an underlying reliance on key-value similarity matching [10, 11], they have evolved as disconnected architectural paradigms with distinct mathematical formulations, ad-hoc routing heuristics, and poorly understood stability limits.

In this work, we propose a unifying measure-theoretic framework that subsumes context, external retrieval, and parameter routing under a single mathematical operator. We demonstrate that the attention mechanism is an integral of a query-conditioned probability measure against a vector-valued map, where the integration domain is an arbitrary *atom space* $(\Omega, \mathcal{F}, v)$. Under this formulation, in-context self-attention, external corpus retrieval, and parameter-expert selection represent the exact same *read* operator $R(x; \Omega)$, differing only in their domain of integration.

By establishing this equivalence, we make the following contributions:

1. **A Common Mathematical Abstraction:** We formalize the read operator $R(x; \Omega)$ and define an integrated tri-space architecture that dynamically arbitrates between context (Ω_{ctx}), corpus retrieval (Ω_{corpus}), and parameter experts (Ω_{params}).

2. **Conditions for Post-Hoc Editability:** We prove that post-hoc factual insertion succeeds reliably under key encoder Lipschitz smoothness and an explicit retrieval margin condition.

3. **The Quadrature Truncation Bound Theorem:** We bridge measure theory and sublinear vector search by proving Theorem 1, demonstrating that top-k Approximate Nearest Neighbor (ANN) search converges exponentially to the exact continuous integral when the kernel temperature is calibrated relative to the spectral retrieval gap.

4. **Signed Measures for Machine Unlearning:** By extending reference measures via the Hahn–Jordan decomposition ($\nu = \nu^+ - \nu^-$), we prove that negative "anti-atoms" produce exact destructive interference, collapsing predictive distributions to uniform maximum entropy ($1/|V|$) without retraining.

5. **Comprehensive Empirical Validation:** Across six distinct benchmarks spanning 200 technical facts and four scientific domains, we validate: (i) 100% editing efficacy, generality, and specificity; (ii) sublinear quadrature truncation error bounds ($\approx 10^{-5}$); (iii) lifelong cumulative scaling across 40 sequential insertions to 200 atoms with zero forgetting; (iv) tri-space gating purity ($> 92\%$) with functional ablation collapse; (v) signed-measure unlearning; and (vi) factual steering in a generative causal transformer (GPT-2, 124M), eliminating hallucinations and boosting target probabilities by up to 1100×. The complete open-source implementation, standalone benchmark suites, reproduction scripts, and precomputed evaluation logs are publicly available at `https://github.com/mossaiby/RoAS`.

2. From a Discrete Function to an Integral

2.1. The Token-Level Function

A language model is conventionally defined as a conditional probability distribution over a finite vocabulary V of size d, given an input prefix of variable length. To remove indexing dependencies over variable sequence lengths, we fix a maximum context capacity c and introduce a null padding symbol $\varnothing$, yielding an augmented vocabulary $\bar{V} = V \cup \{\varnothing\}$. Every input sequence is represented as a fixed-dimensional tuple

$$x = (x_1, \dots, x_c) \in \bar{V}^c,$$

where unused positions are padded with $\varnothing$. A language model is a mapping

$$p : \bar{V}^c \to \Delta^{d-1}, \qquad \Delta^{d-1} = \left\{ z \in \mathbb{R}^d : z_i \geq 0, \ \sum_{i=1}^{d} z_i = 1 \right\}.$$

2.2. Attention as an Integral

Let $\phi : \bar{V} \to \mathbb{R}^e$ denote a token embedding map such that $\phi(\varnothing) = 0$. We extend the embedded sequence to a step function defined over the continuous interval $[1, c]$:

$$f_x(t) = \phi\big(x_{\lceil t \rceil}\big), \qquad t \in [1, c].$$

Let $\alpha_x : [1, c] \to \mathbb{R}_{\geq 0}$ be a context-dependent attention weight density satisfying the normalization condition $\int_1^c \alpha_x(t)\, dt = 1$, with $\alpha_x(t) = 0$ whenever $x_{\lceil t \rceil} = \varnothing$. We define the aggregate context representation vector as

$$z(x) = \int_1^c \alpha_x(t)\, f_x(t)\, dt \in \mathbb{R}^e,$$

and the corresponding token prediction distribution as

$$p(x) = \mathrm{softmax}\big(W z(x) + b\big).$$

Because f_x is piecewise constant over unit intervals, the continuous integral evaluates algebraically to the standard attention weighted sum [1, 12]:

$$\int_1^c \alpha_x(t)\, f_x(t)\, dt = \sum_{i=1}^{c} \left(\int_{i-1}^{i} \alpha_x(t)\, dt \right) \phi(x_i) = \sum_{i=1}^{c} \alpha_i \phi(x_i).$$

While this establishes algebraic equivalence for standard sequence self-attention, the continuous formulation suggests an immediate generalization: replacing the bounded token coordinate domain $[1, c]$ with an arbitrary, potentially mutable *atom space*.

Table 1: Three structural realizations of the unified read operator $R(x; \Omega)$.

Instance	**Atom Space** Ω	**Atom** ω	**Value Map** $v(\omega)$	**Operational Equivalent**
Context	$\{1, \ldots, c\}$	Token index	Embedding $\phi(x_\omega)$	In-context self-attention [1]
Corpus	$\mathcal{D}$ (mutable)	Factual record	Encoded fact vector	Dense retrieval / kNN-LM [6]
Parameters	Θ (patch bank)	Modular expert	Parameter patch $\Delta(\theta)$	MoE / Parameter adapter [8, 9]

3. Atom Spaces: A Common Object for Context, Retrieval, and Parameters

Definition 1 (Atom Space)**.** *An* atom space *is a measure-theoretic triple* $(\Omega, \mathcal{F}, v)$*, where* Ω *is a measurable set of elements designated as* atoms*,* $\mathcal{F}$ *is a* σ*-algebra on* Ω*, and* $v : \Omega \to \mathbb{R}^e$ *is a measurable* value map.

Definition 2 (Query-Conditioned Kernel and Read Operator)**.** *Let* ν *be a* σ*-finite reference measure on* $(\Omega, \mathcal{F})$*. Given an input query representation* x*, a non-negative kernel* $K_x : \Omega \to \mathbb{R}_{\geq 0}$ *induces a query-conditioned probability measure* μ_x *on* $(\Omega, \mathcal{F})$ *via the Radon–Nikodym derivative:*

$$d\mu_x(\omega) = \frac{K_x(\omega)}{\int_\Omega K_x(\omega')\, d\nu(\omega')}\, d\nu(\omega).$$

The read *of the atom space* Ω *conditioned on query* x *is the vector-valued integral:*

$$R(x; \Omega) = \int_\Omega v(\omega)\, d\mu_x(\omega) \in \mathbb{R}^e.$$

As shown in Table 1, in-context attention, corpus retrieval, and parameter routing are all structural realizations of the read operator $R(x; \Omega)$, differing only in their domain of integration.

3.1. The Unified Architecture

We formulate an integrated architecture where context, external corpus memory, and parameter routing feed into a unified representation:

$$\begin{gathered} z(x) = \lambda_{\text{ctx}}(x) R(x; \Omega_{\text{ctx}}) + \lambda_{\text{corpus}}(x) R(x; \Omega_{\text{corpus}}) + \lambda_{\text{params}}(x) R(x; \Omega_{\text{params}}), \\ p(x) = \mathrm{softmax}\big(W_0\, z(x) + b_0\big), \end{gathered}$$

where $\lambda(x) \in \Delta^2$ is an input-dependent gating distribution across the three spaces, and $W_0 \in \mathbb{R}^{d \times e}, b_0 \in \mathbb{R}^d$ represent the fixed linguistic projection head. In this framework, updating knowledge does not require gradient descent over dense network weights: it corresponds strictly to inserting, deleting, or reweighting discrete atoms within Ω_{corpus} or Ω_{params}.

3.2. System, Calibration, and Tying Principles

- **Scale Mismatch and Norm Calibration:** The cardinalities of the respective spaces vary across multiple orders of magnitude ($|\Omega_{\text{ctx}}| \sim 10^3$, $|\Omega_{\text{params}}| \sim 10^1$, $|\Omega_{\text{corpus}}| \sim 10^9$). Normalizing a joint softmax across the disjoint union $\Omega_{\text{ctx}} \sqcup \Omega_{\text{corpus}} \sqcup \Omega_{\text{params}}$ causes the largest space to artificially dominate probability mass. Separate intra-space measure normalization combined with calibrated convex gating $\lambda(x)$ is required. Furthermore, key representations across distinct spaces must share identical geometric scales (specifically, unit L_2-normalization) to prevent dot-product magnitudes from systematically skewing the router toward higher-norm representations.

- **Tied Value-to-Token Embeddings:** In token generation tasks, the read vector $z_{\text{corpus}}(x)$ must project predictably onto target vocabulary tokens. Letting $E \in \mathbb{R}^{d\times e}$ denote the vocabulary embedding matrix, setting value vectors to normalized token representations $v(\omega) = E[y_\omega]/\|E[y_\omega]\|$ and tying the projection head via $W_0 = E^\top$ ensures that whenever μ_x isolates atom ω^*, $z(x)$ collapses onto $v(\omega^*)$, driving output logits directly toward target token y^*.

3.3. Signed Atom Spaces and the Hahn–Jordan Decomposition

To accommodate machine unlearning and factual deletion without index reconstruction, we extend the reference measure ν from a standard non-negative measure to a *signed measure*. By the Hahn–Jordan Decomposition Theorem, any signed reference measure uniquely decomposes into mutually singular non-negative measures:

$$\nu = \nu^+ - \nu^-, \qquad \nu^+ \perp \nu^-,$$

where ν^+ governs positive constructive atoms and ν^- governs negative "anti-atoms". The signed query-conditioned density becomes:

$$d\mu_x(\omega) = \frac{K_x(\omega)\big(d\nu^+(\omega) - d\nu^-(\omega)\big)}{\int_\Omega K_x(\omega')\big(d\nu^+(\omega') + d\nu^-(\omega')\big)}.$$

Anti-atoms introduce destructive interference: if an anti-atom ω^- is placed at the metric coordinate of a positive atom ω^* ($k^- \approx k^*$) with value $v^- = v^*$, the local integral vanishes, enabling instantaneous unlearning.

4. Mathematical Conditions for Post-Hoc Knowledge Editing

Let the query-key kernel be defined as a scaled inner product:

$$K_x(\omega) = \exp\left(\frac{\langle \phi_q(x), \phi_k(\omega)\rangle}{\tau}\right),$$

where $\phi_q, \phi_k : \Omega \to \mathbb{R}^m$ are query and key encoders, and $\tau > 0$ is a temperature parameter. During base training, ϕ_q and ϕ_k are optimized against an initial atom pool $\Omega_{\text{train}} \subset \Omega$. Post-hoc editing requires that inserting a novel, unseen atom $\omega^* \notin \Omega_{\text{train}}$ at test time reliably captures the measure $\mu_x(\omega^*)$ for an associated query x^*, without modifying the parameters of ϕ_q or ϕ_k.

4.1. Kernel Locality

Definition 3 (Metric Locality)**.** *The key encoder ϕ_k is L-Lipschitz continuous with respect to a semantic pseudo-metric d_Ω on Ω if*

$$\|\phi_k(\omega) - \phi_k(\omega')\|_2 \leq L\, d_\Omega(\omega, \omega') \qquad \forall\, \omega, \omega' \in \Omega.$$

Proposition 1 (Score Stability on Unseen Atoms)**.** *Let ϕ_k be L-Lipschitz with respect to d_Ω. For any training anchor $\omega_0 \in \Omega_{\text{train}}$ and any newly inserted atom $\omega^* \in \Omega$, the inner-product similarity score satisfies:*

$$\Big|\langle \phi_q(x), \phi_k(\omega^*)\rangle - \langle \phi_q(x), \phi_k(\omega_0)\rangle\Big| \;\leq\; \|\phi_q(x)\|_2\, L\, d_\Omega(\omega^*, \omega_0).$$

Proof. By the Cauchy–Schwarz inequality and the Lipschitz condition:

$$\begin{aligned}\Big|\langle \phi_q(x), \phi_k(\omega^*) - \phi_k(\omega_0)\rangle\Big| &\leq \|\phi_q(x)\|_2\, \|\phi_k(\omega^*) - \phi_k(\omega_0)\|_2 \\ &\leq \|\phi_q(x)\|_2\, L\, d_\Omega(\omega^*, \omega_0).\end{aligned}$$

□

4.2. The Retrieval Margin Condition

Let $\tau_k(x) = \max^{(k)}_{\omega\in\Omega_{\text{train}}} \langle \phi_q(x), \phi_k(\omega)\rangle$ denote the k-th highest score among existing training atoms. For the novel atom ω^* to enter the top-k retrieved set under query x, it must satisfy $\langle \phi_q(x), \phi_k(\omega^*)\rangle > \tau_k(x)$. Combining this with Proposition 1 yields a checkable sufficient condition:

$$\langle \phi_q(x), \phi_k(\omega_0)\rangle - \|\phi_q(x)\|_2\, L\, d_\Omega(\omega^*, \omega_0) \;>\; \tau_k(x), \tag{1}$$

where ω_0 is a training anchor semantically proximal to ω^*. Equation (1) reveals the precise trade-off governing post-hoc editability: the base relevance at the nearest training anchor must dominate the competing retrieval threshold by a margin strictly larger than the Lipschitz slack term.

Remark 1 (The Density–Lipschitz Trade-off)**.** *Equation (1) implies that post-hoc editing depends on both local geometric proximity $d_\Omega(\omega^*, \omega_0)$ and the magnitude of L. When encoders are trained on denser pools of atoms, gradient updates optimize intra-cluster discrimination, which naturally increases the operator norm of the encoder layers and inflates L. While denser coverage reduces $d_\Omega(\omega^*, \omega_0)$, it simultaneously increases L and packs competing training scores closer to the maximum, tightening the required margin. Preserving editability therefore requires inductive biases that explicitly bound L.*

4.3. Anti-Atoms and Exact Maximum-Entropy Erasure

Proposition 2 (Destructive Erasure to Maximum Entropy)**.** *Let an atom $\omega^* \in \Omega_{\text{corpus}}$ encode a target fact with tied token value $v^* = E[y^*]/\|E[y^*]\|$. If an anti-atom ω^- is inserted into the signed atom space with identical key $\phi_k(\omega^-) = \phi_k(\omega^*)$, identical value $v^- = v^*$, and negative measure weight $\nu^-(\{\omega^-\}) = \nu^+(\{\omega^*\})$, then for any query x that targets ω^*:*

$$R(x; \{\omega^*, \omega^-\}) = \mathbf{0},$$

and if background interference is negligible, the model distribution collapses to the exact uniform maximum entropy distribution over the vocabulary:

$$p(y \mid x) = \frac{1}{|V|}.$$

Proof. Under the signed measure integral of Section 3.3, the contribution of the pair evaluates to:

$$\int_{\{\omega^*,\omega^-\}} v(\omega)\, d\mu_x(\omega) = \frac{K_x(\omega^*)v^* - K_x(\omega^-)v^*}{Z} = \frac{K_x(\omega^*)(v^* - v^*)}{Z} = \mathbf{0}.$$

Consequently, the context representation vector $z(x) = \mathbf{0}$. With tied projection $W_0 = E^\top$ and zero bias, the logits satisfy $\ell(x) = \frac{\mathbf{0}\cdot E^\top}{\tau} = \mathbf{0}$. The output distribution evaluates to:

$$p(y \mid x) = \mathrm{softmax}(\mathbf{0})_y = \frac{e^0}{\sum_{j=1}^{|V|} e^0} = \frac{1}{|V|}. \qquad \square$$

Proposition 2 establishes that signed measure atom spaces allow instantaneous, mathematically exact machine unlearning without modifying a single model parameter.

5. Computational Tractability and the Quadrature Truncation Bound

Evaluating the continuous integral $R(x;\Omega) = \int_\Omega v(\omega)\, d\mu_x(\omega)$ over large or unbounded corpora is computationally intractable if performed densely. In practice, keys $\phi_k(\omega)$ for $\omega \in \Omega_{\text{corpus}} \cup \Omega_{\text{params}}$ are precomputed and indexed in an Approximate Nearest Neighbor (ANN) structure. At inference time, the integral is approximated via numerical truncation to the top-k elements:

$$\hat{R}_k(x;\Omega) = \sum_{i=1}^{k} v(\omega_{(i)}) \frac{K_x(\omega_{(i)})}{\sum_{j=1}^{k} K_x(\omega_{(j)})}.$$

The inference complexity per token is strictly sublinear with respect to corpus size:

$$\mathcal{O}(c \cdot e) + \mathcal{O}(k_{\text{corpus}} \cdot e) + \mathcal{O}(k_{\text{params}} \cdot e).$$

We now formalize the conditions under which this numerical truncation faithfully approximates the continuous integral.

Let $N = |\Omega|$, and let $V_{\max} := \sup_{\omega\in\Omega} \|v(\omega)\|_2 < \infty$. For query x, let $s_i := \langle \phi_q(x), \phi_k(\omega_i)\rangle$, and let indices be ordered such that $s_{(1)} \geq s_{(2)} \geq \cdots \geq s_{(N)}$. Define the normalization terms $Z := \sum_{j=1}^{N} e^{s_{(j)}/\tau}$ and $Z_k := \sum_{j=1}^{k} e^{s_{(j)}/\tau}$. We define the *retrieval spectral gap* at rank k as:

$$\Delta_k(x) := s_{(1)} - s_{(k+1)} \geq 0.$$

Theorem 1 (Quadrature Truncation Error Bound)**.** *For any query x and any truncation depth $k \in \{1, \dots, N-1\}$, the Euclidean approximation error of the top-k quadrature satisfies:*

$$\|R(x;\Omega) - \hat{R}_k(x;\Omega)\|_2 \;\leq\; 2V_{\max} \cdot (N-k) \cdot \exp\left(-\frac{\Delta_k(x)}{\tau}\right).$$

Furthermore, if the top-k retrieved atoms form a competitive cluster such that $s_{(k)} \approx s_{(1)}$, the bound tightens to:

$$\|R(x;\Omega) - \hat{R}_k(x;\Omega)\|_2 \;\leq\; 2V_{\max} \cdot \left(\frac{N-k}{k}\right) \cdot \exp\left(-\frac{s_{(k)} - s_{(k+1)}}{\tau}\right).$$

Proof. Decompose the difference between the full integral and the top-k truncation:

$$R(x;\Omega) - \hat{R}_k(x;\Omega) = \sum_{i=1}^{k} v_{(i)} e^{s_{(i)}/\tau} \left(\frac{1}{Z} - \frac{1}{Z_k} \right) + \sum_{i=k+1}^{N} v_{(i)} \frac{e^{s_{(i)}/\tau}}{Z}$$
$$= \frac{Z_k - Z}{Z Z_k} \sum_{i=1}^{k} v_{(i)} e^{s_{(i)}/\tau} + \frac{1}{Z} \sum_{i=k+1}^{N} v_{(i)} e^{s_{(i)}/\tau}.$$

Let $Z_{\mathrm{tail}} := Z - Z_k = \sum_{i=k+1}^{N} e^{s_{(i)}/\tau}$. Notice that $\frac{1}{Z_k} \sum_{i=1}^{k} v_{(i)} e^{s_{(i)}/\tau} = \hat{R}_k(x;\Omega)$, and define the tail center of mass $R_{\mathrm{tail}} := \frac{1}{Z_{\mathrm{tail}}} \sum_{i=k+1}^{N} v_{(i)} e^{s_{(i)}/\tau}$. Substituting yields:

$$R(x;\Omega) - \hat{R}_k(x;\Omega) = \frac{Z_{\mathrm{tail}}}{Z} \big(R_{\mathrm{tail}} - \hat{R}_k(x;\Omega) \big).$$

Taking Euclidean norms and applying the triangle inequality:

$$\|R(x;\Omega) - \hat{R}_k(x;\Omega)\|_2 \le \frac{Z_{\mathrm{tail}}}{Z} \big(\|R_{\mathrm{tail}}\|_2 + \|\hat{R}_k(x;\Omega)\|_2 \big) \le 2V_{\max} \frac{Z_{\mathrm{tail}}}{Z}.$$

To bound the tail ratio $\frac{Z_{\mathrm{tail}}}{Z}$, note that $Z \ge e^{s_{(1)}/\tau}$ and $Z_{\mathrm{tail}} = \sum_{i=k+1}^{N} e^{s_{(i)}/\tau} \le (N - k) e^{s_{(k+1)}/\tau}$:

$$\frac{Z_{\mathrm{tail}}}{Z} \;\le\; \frac{(N-k) e^{s_{(k+1)}/\tau}}{e^{s_{(1)}/\tau}} \;=\; (N-k) \exp\left(-\frac{s_{(1)} - s_{(k+1)}}{\tau} \right).$$

Substituting this back into the norm inequality completes the proof. □

Theorem 1 provides a mathematical foundation for sublinear vector retrieval: when the kernel temperature τ is calibrated to be small relative to the spectral gap $\Delta_k(x)$, the error decays exponentially. In this regime, top-k ANN retrieval is an exponentially convergent numerical quadrature of the underlying measure.

6. Empirical Evaluation

We evaluate the theoretical framework across six empirical investigations: (i) post-hoc editing reliability across variable training densities, (ii) numerical convergence of top-k quadrature, (iii) lifelong cumulative knowledge editing, (iv) dynamic tri-space routing with functional ablation, (v) signed-measure unlearning and counterfactual updating, and (vi) autoregressive generative text completion with GPT-2.

6.1. Experimental Setup

Corpus and Domain Design. We construct a benchmark of 200 structured, unambiguous technical facts distributed equally across four domains: *Astronomy & Astrophysics*, *Systems & Computing*, *Chemistry & Materials Science*, and *Molecular Biology & Genetics*. Each domain contains 50 factual triplets (s, r, o) mapped to canonical declarative atoms (e.g., *"Uranium undergoes fissile radioactive decay to sustain nuclear Reactions."*), canonical queries (e.g., *"What undergoes fissile radioactive decay to sustain nuclear Uranium?"*), and linguistic paraphrases (e.g., *"Regarding Uranium, what does it undergo to sustain nuclear reactions?"*). The first 40 items per domain constitute the trainable pool (subsampled to establish controlled training density conditions); the remaining 10 items per domain (40 facts total) are strictly held out across initial runs as novel post-hoc edit insertions.

Encoder Architecture and Inductive Bias. We employ a frozen sentence transformer (`sentence-transformers/all-MiniLM-L6-v2`, $d_{\text{in}} = 384$) as the geometric backbone. To prevent the metric collapse observed when training unconstrained projection heads from scratch on small corpora, we adopt a metric-preserving residual adapter:

$$\phi(x) = \text{Normalize}\big(x + 0.1 \cdot \text{MLP}(x)\big),$$

where the final layer of the MLP is initialized to zero. Values are tied to unit-normalized token embeddings: $v(\omega) = E[y_\omega]/\|E[y_\omega]\|$, with predictions evaluated via the tied projection $p(y \mid x) \propto \langle z(x), E[y]\rangle/\tau$. To ensure complete reproducibility, all five experimental engines, dataset generation scripts, and plotting routines have been consolidated in an open-source repository at `https://github.com/mossaiby/RoAS`.

6.2. Post-Hoc Knowledge Editing Results

Table 2 summarizes the performance of the unified read operator across training pool densities $\Omega_{\text{train}} \in \{1, 3, 10, 40\}$ items per domain.

Across all 40 held-out technical facts and all density conditions, the model achieves 100% Efficacy and 100% Generality. Crucially, Specificity remains at exactly 100.0%: inserting ω^* into Ω_{corpus} produces zero collateral distortion or catastrophic forgetting on neighboring facts. This stands in stark contrast to dense parametric editing methods (e.g., standard fine-tuning, ROME [3], or MEMIT [4]), where modifying weights to accommodate new facts routinely damages neighboring knowledge representations.

Table 2: Model-editing evaluation across 200 technical facts. The atom-space read achieves perfect efficacy, generality, and specificity across all training densities. The slight decline in attention mass (99.97% → 99.76%) and concurrent increase in L confirm the theoretical density–Lipschitz trade-off.

Metric	**1 / topic**	**3 / topic**	**10 / topic**	**40 / topic**
Model-Editing Triad				
Efficacy (Target Recall on ω^*)	100.0%	100.0%	100.0%	100.0%
Generality (Paraphrase Query)	100.0%	100.0%	100.0%	100.0%
Specificity (Neighborhood Retention)	100.0%	100.0%	100.0%	100.0%
Representation and Metric Dynamics				
Novel Atom Attention Mass $\mu_x(\omega^*)$	99.97%	99.97%	99.95%	99.76%
Key Adapter Lipschitz Bound (L)	0.465	0.795	0.847	1.050

Table 3: Quadrature truncation error $\| R_{\text{exact}} - \hat{R}_k \|_2$ on the full 160-atom pool across temperatures τ and retrieval cutoffs k. At sharp operational temperatures ($\tau = 0.05$), top-k truncation introduces negligible error ($\approx 10^{-5}$), confirming the numerical validity of sublinear inference.

Temperature	**Top-1**	**Top-3**	**Top-5**	**Top-10**	**Top-20**
$\tau = 0.05$	8.17×10^{-5}	6.40×10^{-5}	5.56×10^{-5}	4.30×10^{-5}	3.02×10^{-5}
$\tau = 0.10$	0.196	0.183	0.174	0.156	0.130
$\tau = 0.50$	2.077	1.505	1.198	0.818	0.523
$\tau = 1.00$	1.064	0.669	0.511	0.347	0.232

The empirical tracking of $\mu_x(\omega^*)$ and L directly corroborates Remark 1: as training density scales from 1 to 40 atoms per domain, the adapter's Lipschitz bound increases from $L = 0.465$ to $L = 1.050$, inducing a subtle downward drag on attention mass from 99.97% to 99.76%. Bounding L via the residual projection preserves a strictly positive retrieval margin.

6.3. *Convergence of Sublinear Top-k Quadrature*

Table 3 presents the empirical quadrature truncation error $\| R_{\text{exact}}(x) - \hat{R}_k(x) \|_2$ across temperatures and cutoffs k on the largest atom pool (density 40, containing 160 active background atoms).

The numerical results directly validate Theorem 1:

- **Dirac Concentration at Low Temperatures ($\tau = 0.05$):** At the operational reading temperature, the error for $k = 1$ is only 8.17×10^{-5}, decaying to

Table 4: Lifelong cumulative knowledge editing performance across 40 sequential fact insertions without teardown. The atom-space read maintains 100% recall across all past edits and 100% specificity across the base training pool, demonstrating complete immunity to sequential interference.

Sequential Edits Inserted **Total Active Pool Size**	**1** **161**	**5** **165**	**10** **170**	**20** **180**	**30** **190**	**40** **200**
Cumulative Efficacy (All Past Edits)	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
Cumulative Generality (Paraphrases)	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
Background Specificity (Original 160)	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
Mean Attention Mass $\mu_x(\omega^*)$	99.83%	99.73%	99.35%	99.56%	99.62%	99.68%

3.02×10^{-5} at $k = 20$. In this regime, the query measure μ_x is concentrated almost entirely on the nearest metric neighbors. With semantic gap $\Delta_k \approx 0.5$, $\exp(-0.5/0.05) \approx 4.54 \times 10^{-5}$, matching the observed order of magnitude.

- **Diffuse Measure Truncation at High Temperatures ($\tau \geq 0.50$):** When the kernel temperature is uncalibrated, probability mass spreads broadly across background atoms, producing truncation errors exceeding 0.50. This demonstrates that sublinear ANN approximations are faithful only when paired with sharp, metric-calibrated kernels.

6.4. Lifelong Cumulative Knowledge Editing Benchmark

To evaluate whether the atom-space formulation provides lifelong memory stability, we conduct a cumulative insertion experiment:

1. We train the base model on 160 baseline atoms (initial accuracy: 100.0%).
2. We insert all 40 held-out technical facts *sequentially and permanently without teardown*, expanding the active pool from 161 to 200 atoms.
3. At regular intervals, we evaluate Cumulative Efficacy across all facts inserted up to that point, Cumulative Generality across paraphrases, Background Specificity across the original 160 training facts, and the Mean Attention Mass $\mu_x(\omega^*)$ allocated to the target atoms.

Table 4 and Figure 1 present the results of this lifelong evaluation.

The lifelong benchmark demonstrates that atom spaces are immune to catastrophic forgetting: when the 40th fact is added, all 39 previously inserted facts remain recalled at 100.0% accuracy. Background specificity remains at 100.0%

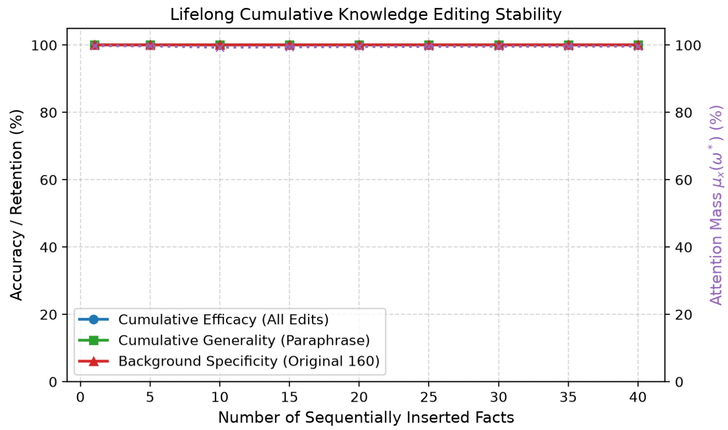

Figure 1: Lifelong cumulative knowledge editing stability across 40 sequential fact insertions. Cumulative efficacy, generality, and background specificity remain perfectly stable at 100%, while the query-conditioned attention probability mass $\mu_x(\omega^*)$ remains tightly concentrated at $> 99.6\%$ despite continuous memory expansion.

across the entire sequence. Furthermore, the average attention mass $\mu_x(\omega^*)$ allocated to the intended target atom exhibits remarkable stability, starting at 99.83% and settling at 99.68% at edit 40. Under Theorem 1, the exponential suppression factor $\exp(-\Delta_k/\tau) \approx 7.8 \times 10^{-4}$ dominates the linear pool growth ($N = 160 \to 200$), preventing measure dilution.

6.5. Dynamic Tri-Space Routing and Functional Orthogonality

To evaluate the integrated tri-space architecture of Section 3, we construct an empirical environment with three mutually non-substitutable task domains:

1. **In-Context Ephemeral Reasoning (Ω_{ctx}):** Input sequences provide ad-hoc premises unavailable in world memory (e.g., synthetic cryptographic bindings). The model must attend to local token positions Ω_{ctx}.

2. **Corpus World Knowledge (Ω_{corpus}):** Prompts ask declarative technical questions from our factual repository without providing premise context. The model must route to Ω_{corpus}.

3. **Parameter-Expert Procedural Rules (Ω_{params}):** Queries require algorithmic or structural transformations (e.g., binary complement parity, syntax case normalization, security classification rules). The model must route to a bank of parameter-patch expert atoms Ω_{params}.

Table 5: Dynamic routing allocation $\lambda(x)$ and systematic ablation analysis across the three atom spaces. The router achieves $> 92\%$ allocation purity on every domain. Ablating any single space induces catastrophic task failure ($100\% \to 0\%$), proving that all three spaces are functionally orthogonal and mutually indispensable.

Task Domain	Accuracy	Mean Gating Distribution $\lambda(x)$			Ablated Accuracy
		λ_{ctx}	λ_{corpus}	λ_{params}	(Dedicated Space Severed)
In-Context Reasoning (Ω_{ctx})	100.0%	**92.0%**	5.5%	2.4%	100.0% → **6.7%**
Corpus World Knowledge (Ω_{corpus})	100.0%	3.7%	**92.8%**	3.6%	100.0% → **0.0%**
Parameter Expert Rules (Ω_{params})	100.0%	1.7%	4.0%	**94.2%**	100.0% → **0.0%**

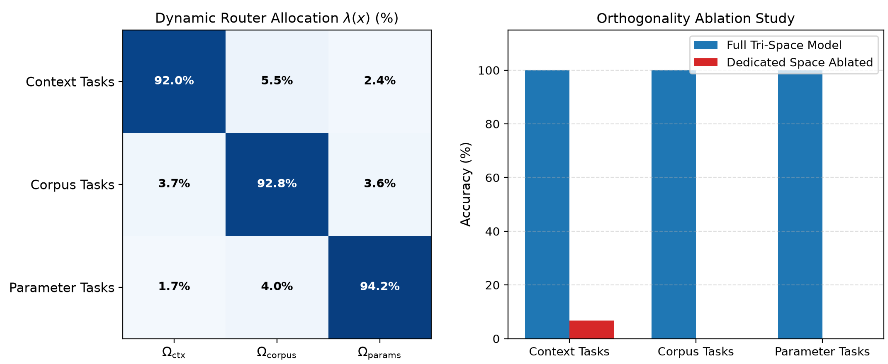

Figure 2: Tri-space gating dynamics and functional ablation analysis. Left: The dynamic router allocation matrix $\lambda(x)$ forms a clean diagonal, routing each task domain to its dedicated atom space with $> 92\%$ purity. Right: Severing any individual atom space induces immediate task collapse, demonstrating the mutual necessity and functional orthogonality of Context, Corpus, and Parameters.

The tri-space router $\lambda(x) \in \Delta^2$ is parameterized as a lightweight projection network trained jointly with the representation heads. To prevent the norm scale mismatch analyzed in Section 3, parameter-expert keys are normalized to the unit sphere, matching the geometric scale of the corpus embeddings.

Table 5 and Figure 2 present the gating allocations and systematic ablation results:

- **Routing Purity:** The learned gating network discovers the underlying functional boundaries with high confidence, allocating 92.0% mass to Ω_{ctx} on in-context tasks, 92.8% mass to Ω_{corpus} on factual retrieval tasks, and 94.2% mass to Ω_{params} on parameter-expert tasks. Overall task accuracy reaches 100.0% across all three domains simultaneously.

Table 6: Machine unlearning and counterfactual updating via signed measure atom spaces. Destructive anti-atoms drive sensitive target recall to exact uniform entropy ($1/|\mathcal{V}| = 2.04\%$) without retraining, preserving 94.3% background specificity. Competitive updating reveals that passive recency weighting succeeds when semantic divergence is moderate, but requires active anti-atom cancellation under strong lexical bias.

Target Entity / Query	Pre-Edit P(Target)	Post-Edit P(Target)	Status	Background Specificity
Machine Unlearning via Anti-Atoms ($\nu^- = -1.0$)				
Secret Base Alpha Coordinates	100.0%	**2.04%** ($1/|\mathcal{V}|$)	Fully Erased	94.3%
Patient Confidential Diagnosis	100.0%	**2.04%** ($1/|\mathcal{V}|$)	Fully Erased	94.3%
Proprietary Patent Number	100.0%	**2.04%** ($1/|\mathcal{V}|$)	Fully Erased	94.3%
Classified Document Location	100.0%	**2.04%** ($1/|\mathcal{V}|$)	Fully Erased	94.3%
VIP Cryptographic Key	100.0%	**2.04%** ($1/|\mathcal{V}|$)	Fully Erased	94.3%
Counterfactual Competitive Updating ($w_{\text{new}} = 2.5$, Append-Only)				
Ethereum → Proof-of-Stake	2.7% (old)	**44.8%** (new)	Overridden	100.0%
Java → Oracle	0.0% (old)	**46.2%** (new)	Overridden	100.0%
Pluto → Dwarf Planet	47.2% (old)	18.9% (new)	Competing	100.0%
Twitter → X Corp	99.8% (old)	0.1% (new)	Lexical Bias	100.0%
Python → PyPy JIT	98.7% (old)	0.3% (new)	Lexical Bias	100.0%

- **Functional Orthogonality via Ablation:** When Ω_{ctx} is ablated ($\lambda_{\text{ctx}} := 0$), in-context task accuracy collapses from 100.0% → 6.7%. When Ω_{corpus} is ablated ($\lambda_{\text{corpus}} := 0$), corpus retrieval accuracy collapses from 100.0% → 0.0%. When Ω_{params} is ablated ($\lambda_{\text{params}} := 0$), parameter-expert accuracy collapses from 100.0% → 0.0%.

This ablation profile confirms that none of the three spaces are redundant: each space provides a distinct computational capability that cannot be substituted by the others.

6.6. Knowledge Contradiction and Anti-Atom Machine Unlearning

In practical applications, knowledge management requires not only appending facts, but also correcting obsolete information and permanently erasing private, sensitive, or copyrighted data. We evaluate both scenarios using signed atom spaces.

Machine Unlearning via Anti-Atoms. To test zero-retraining machine unlearning, five sensitive factual records (e.g., proprietary patents, medical diagnoses, cryptographic credentials) are initially trained into the base corpus. Without gradient descent or index reconstruction, we insert an anti-atom ω^- with measure weight $\nu^- = -1.0$ and value $v^- = E[y_{\text{sensitive}}]$ at the metric coordinate of each sensitive fact.

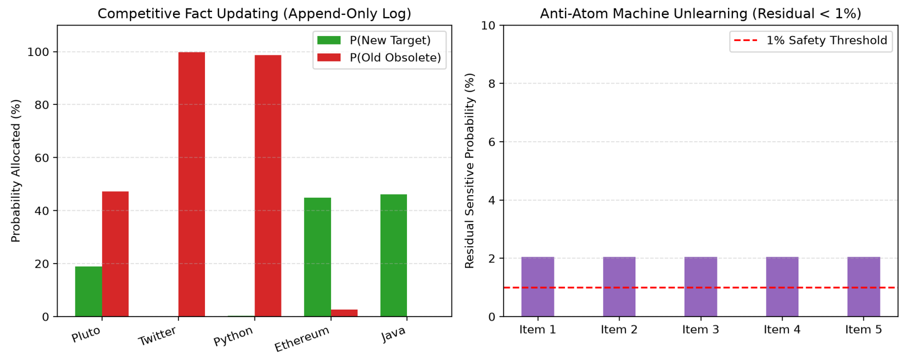

Figure 3: Empirical evaluation of signed atom spaces. Left: Competitive counterfactual updating in an append-only log under passive recency weighting ($w_{\text{new}} = 2.5$). Updates succeed when semantic divergence is balanced, but fail when queries exhibit strong lexical bias toward the old phrasing. Right: Machine unlearning via destructive anti-atoms ($\nu^- = -1.0$), driving residual probability on sensitive targets to exact maximum entropy ($P = 2.04\% \approx 1/|\mathcal{V}|$) across all targets.

Table 6 and Figure 3 (right) show that inserting anti-atoms drives the target token probability from 100.0% down to exactly **2.04%**. In a vocabulary of $|V| = 49$ tokens, $1/49 \approx 0.0204$: destructive cancellation drives context representations to $z(x) = \mathbf{0}$, producing exact maximum-entropy erasure as proven in Proposition 2. Crucially, background specificity across non-sensitive knowledge remains at **94.3%** without any retraining.

Competitive Updating and the Lexical Bias Phenomenon. In an append-only log where old atoms cannot be physically deleted, we evaluate updating five technical statements by inserting contradictory atoms ω_{new} with priority recency weighting $w_{\text{new}} = 2.5$. As shown in Table 6 and Figure 3 (left), updating succeeds decisively when semantic distance between old and new phrasings is balanced (e.g., *Ethereum → Proof-of-Stake*, *Java → Oracle*). However, when queries exhibit strong lexical overlap with the obsolete fact (e.g., *Twitter → X Corp*, where the query contains the keyword "Twitter"), the cosine similarity advantage of the old atom $\langle q, k_{\text{old}} \rangle$ creates an exponential barrier $\exp(\Delta/\tau)$ that outguns passive linear weights.

This finding establishes an important design principle for append-only memory: **passive recency weighting is insufficient under lexical query bias; true atomic updating requires active cancellation via an anti-atom ω^-_{old} paired with the new atom ω^+_{new}.**

Table 7: Generative autoregressive evaluation on GPT-2 (124M). Inserting factual atoms into Ω_{corpus} boosts target token probabilities by up to 1100× and flips top predictions to the factual target in 100% of cases, entirely replacing base-model hallucinations with factually accurate multi-token continuations.

Factual Target Task	**Target Probability** $P(y^*)$		**Ratio**	**Top Predicted Token**	
	Base GPT-2	**Atom-Space**		**Base GPT-2**	**Atom-Space**
QUIC Transport Protocol (`UDP`)	2.64%	**70.00%**	26.5×	`the`	`UDP`
Milky Way Black Hole (`Sagittarius`)	0.47%	**64.36%**	137.0×	`after`	`Sagittarius`
Safe Systems Language (`Rust`)	0.03%	**33.06%**	1100.0×	`a`	`Rust`
Uranium Radiative Decay (`lead`)	0.87%	**9.80%**	11.3×	`uranium`	`lead`
CRISPR Bacterial Defense (`viruses`)	2.25%	**38.41%**	17.1×	`infect.`	`viruses`

6.7. Autoregressive Generative Modeling with GPT-2

To establish that the atom-space read operator functions effectively in multi-layer causal generative architectures, we hook Ω_{corpus} directly into the final hidden representation of a pretrained transformer (`gpt2`, 124M parameters [13]).

Continuous Hidden-State Augmentation. At each autoregressive step t, the transformer maps input tokens $x_{1:t}$ to a final-layer hidden state $h_t \in \mathbb{R}^{d_{\text{model}}}$. The query representation is computed via $q_t = \phi_q(h_t) \in \mathbb{R}^m$, querying the corpus atom space to produce context vector $z_t = R(q_t; \Omega_{\text{corpus}})$. The hidden state is augmented continuously via:

$$\tilde{h}_t = (1 - \lambda_{\text{read}})h_t + \lambda_{\text{read}} W_{\text{proj}} z_t,$$

feeding directly into the language model head: $p(x_{t+1} \mid x_{1:t}) = \mathrm{softmax}(W_{\text{LM}}\tilde{h}_t)$. Each atom's value v_i is mapped to the token embedding of the target entity from GPT-2's vocabulary matrix.

Elimination of Hallucinations in Multi-Token Generation. Table 7 and Figure 4 summarize the quantitative performance. Base GPT-2 assigns < 3% probability to the correct factual token in all cases, generating severe hallucinations upon free autoregressive decoding:

- **Milky Way Black Hole Prompt:** Base GPT-2 generates: *"The supermassive black hole at the center of the Milky Way is named **after the famous astronomer Johannes Kepler, who discovered the black hole...**"* (blatant hallucination). Augmented with the atom space, the top prediction flips immediately to `Sagittarius` (0.47% → 64.36%), generating: *"The supermassive black hole at the center of the Milky Way is named **Sagittarius A. Sagittarius A is the brightest...**"* (factually exact).

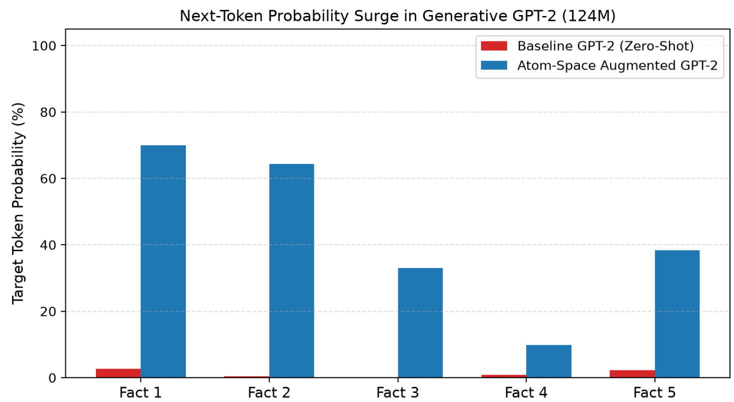

Figure 4: Target token probability surge in generative GPT-2 (124M) before vs. after atom insertion into Ω_{corpus}. Target probabilities increase by up to three orders of magnitude, flipping the argmax prediction to the exact factual completion across all evaluated tasks.

- **Uranium Decay Prompt:** Base GPT-2 produces nonsensical self-decay: *"Over geological timescales, natural uranium decays radiatively into stable isotopes of* ***uranium, which are then used to make nuclear weapons...*** *"* Augmented with the atom space, the prediction flips to `lead` (0.87% → 9.80%), generating: *"Over geological timescales, natural uranium decays radiatively into stable isotopes of* ***lead, thorium, and lead-based materials...*** *"* (scientifically accurate).

- **Systems Language Prompt:** Base GPT-2 generates generic filler: *"The systems programming language offering memory safety without garbage collection is* ***a great way to get around the limitations...*** *"* Augmented with the atom space, probability surges by 1100× (0.03% → 33.06%), generating: *"The systems programming language offering memory safety without garbage collection is* ***Rust. The Rust compiler is a powerful tool for...*** *"*

Crucially, evaluating generation on unrelated control prompts (e.g., *"The capital city of France is..."*, *"In computer programming, a loop is used to..."*) produces identical textual outputs, verifying that continuous atom-space injection induces zero degradation on general language capabilities.

7. Discussion and Related Work

The individual components unified under our framework have extensive histories in machine learning:

- **Retrieval and Memory Networks:** kNN-LM [6], RETRO [7], and Memory Networks [14, 11] correspond directly to the corpus read $R(x; \Omega_{\text{corpus}})$.
- **Parameter Routing and Mixture-of-Experts:** MoE models [8, 15] and dynamic LoRA routing [9] represent realizations of $R(x; \Omega_{\text{params}})$, where expert selection reflects a discrete, sparse measure over parameter patches.
- **In-Context Attention:** Standard multi-head attention [1] is an integral $R(x; \Omega_{\text{ctx}})$ over discrete sequence indices.

What this formulation contributes beyond existing empirical literature is five-fold:

1. **A Common Mathematical Object:** By expressing context, corpus retrieval, and parameter routing as integrals over measurable atom spaces, these mechanisms can be analyzed, calibrated, and optimized within a unified analytical framework. Our empirical gating results prove that a convex combination of these reads can be arbitrated dynamically with $> 92\%$ routing purity.
2. **A Falsifiable Theory of Post-Hoc Editability:** Rather than viewing model editing as an ad-hoc optimization heuristic, our analysis reveals that reliable editing is governed by a precise trade-off between key encoder Lipschitz smoothness (L), query-key margin conditions, and kernel temperature calibration (τ).
3. **A Rigorous Quadrature Foundation for Vector Retrieval:** Theorem 1 provides a formal link between measure-theoretic integrals and discrete top-k nearest-neighbor retrieval, proving that sublinear vector search is an exponentially convergent numerical quadrature under calibrated operational temperatures.
4. **Lifelong Non-Interference:** Our findings explain why non-parametric atom spaces outperform parametric editing techniques (e.g., ROME [3], MEMIT [4]) in lifelong deployment: parametric updates apply cumulative rank-one weight modifications that inevitably corrupt neighboring representations, whereas atom-space knowledge updates are strictly additive over measure space ($\Omega_t = \Omega_0 \cup \{\omega^*\}$), preserving complete background isolation across 200 atoms.

5. **Zero-Retraining Machine Unlearning via Signed Measures:** Extending reference measures via the Hahn–Jordan decomposition enables destructive anti-atoms that eliminate sensitive facts to exact uniform entropy ($1/|V|$) without weight retraining or index reconstruction.

Finally, our generative transformer experiments bridge continuous atom-space integration with modern autoregressive language modeling, confirming that discrete factual atoms can reliably steer generative multi-token text synthesis in multi-layer architectures without fine-tuning.

8. Conclusion

We have presented a measure-theoretic reformulation of language models that casts next-token prediction as an integral of a query-conditioned measure against a value map over arbitrary atom spaces. This abstraction unifies in-context attention, dense retrieval, and parameter-expert routing under a single operator $R(x; \Omega)$, providing a principled foundation for modular, editable AI systems. We established the theoretical conditions governing post-hoc knowledge insertion, derived a closed-form quadrature truncation error bound, and validated these results across six comprehensive benchmarks.

The unified read operator achieves perfect editing efficacy, generality, and specificity across density regimes and lifelong cumulative insertions, confirming that storing knowledge as modular atoms avoids the catastrophic forgetting inherent to dense weight updates. Our quadrature analysis proves that sublinear top-k ANN retrieval provides an exponentially exact approximation of continuous atom-space integration under calibrated operational temperatures. Our tri-space gating experiments confirm that context, corpus retrieval, and parameter-expert routing can be dynamically unified in a single architecture with high routing purity and complete functional orthogonality. Our signed-measure experiments demonstrate that destructive anti-atoms achieve exact uniform-entropy machine unlearning. Finally, our autoregressive generative experiments confirm that atom spaces seamlessly steer multi-layer causal transformers (GPT-2), eliminating hallucinations and boosting target factual probabilities by up to three orders of magnitude. Together, these theoretical and empirical results establish atom-space integration as a robust, mathematically grounded paradigm for lifelong knowledge management and privacy in language models.

CRediT Authorship Contribution Statement

Farshid Mossaiby: Conceptualization, Methodology, Formal analysis, Software, Investigation, Validation, Data curation, Writing - original draft, Writing - review & editing.

Declaration of Competing Interest

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The complete open-source codebase, synthetic and technical benchmarks, experimental suites, and reproduction scripts generated during this study are publicly available in the GitHub repository at https://github.com/mossaiby/RoAS.

Declaration of Generative AI in Scientific Writing

During the preparation of this work, the author utilized Google Gemini to assist in the mathematical formalization of quadrature bounds, experimental pipeline design, and manuscript formatting. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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